User Research Plan

Cybersecurity Mental Models & Everyday Digital Interactions

Research Goal

To understand the **mental models** of users on how they recognise, interpret, and respond to cybersecurity threats in everyday digital interactions, and how they store these interactions in memory. The aim is to identify the **gaps between existing security mechanisms and users' mental models**, and design a solution that helps them spot real threats and know exactly what to do *before*, *during*, and *after* a threat occurs.

Objectives & Research Questions

OBJECTIVE 1 Mental Representation of Threats

How do everyday users mentally picture cybersecurity threats?

1. If a friend told you their phone had been hacked, what would you guess went wrong on their end?
2. Have you ever received a suspicious message and been unsure whether it was real or fake? What made it feel uncertain?
3. Walk me through what you think actually happens when someone falls for a phishing email.

OBJECTIVE 2 Before, During & After a Threat

What does a user actually do at each stage of a security threat, before they spot it (awareness), during the moment they face it (decision), and after something goes wrong (recovery)?

1. Tell me about a time you came across something online that felt suspicious or made you uneasy. Walk me through exactly what you did, from the moment you noticed it.

2. Before that moment happened, was there anything you were regularly doing to keep yourself safe online?
3. In that moment, what options did you feel you had? What did you end up doing, and why that?
4. What did you do after, in the next few minutes, or that same day?
5. Did you ever feel like the situation was fully resolved?

OBJECTIVE 3 The Intention–Behaviour Gap

Why do users who worry about their online safety still fall into cyber attacks?

1. Is there something you know you probably should do for your online safety, like changing a password or updating an app, that you keep postponing? If yes, what has been getting in the way?

OBJECTIVE 4 Attention & Response to Warnings

When a user sees a warning, alert, or notification about a security risk, what do they actually notice? What makes them pay attention, or not?

1. Have you ever dismissed a warning without reading it, and then wondered later if you should have paid attention to it? Tell me about that.
2. Walk me through a typical morning going through your phone, messages, emails, notifications. What does that usually look like?
3. Are there certain times of day or situations, when you are tired, in a hurry, or distracted, when you are more likely to just quickly tap through things without really looking?
4. Are there particular types of messages, delivery updates, bank alerts, prize notifications, password resets, that you tend to react to faster or differently than others?
5. Has there ever been a message or website that looked completely real to you, but turned out to be fake? What made it seem so convincing?
6. When using a new app or signing up for something online, how much attention do you pay to what you are agreeing to, or what the app is asking to access?

OBJECTIVE 5 Sources of Security Knowledge

Where do users actually get their knowledge about cybersecurity? What kind of learning actually changes how they behave?

1. Think about the last time you learned something new about staying safe online. Where did it come from, and what made it stick with you?

2. If you had to explain to a younger sibling or a parent how to spot a fake message, what would you tell them?

Alignment with Literature

Each objective maps directly to a gap or theme identified in the literature review:

- **Mental models & schema theory** (Obj. 1), users hold an inaccurate perspective of threats, leading to poor recognition.

- **Staged threat response** (Obj. 2), Incident-response studies reveal different cognitive patterns that occur at each stage.

- **Intention–behaviour gap** (Obj. 3), Self-reported concern over privacy does not predict change in behaviour.

- **Warning fatigue & habituation** (Obj. 4), repeated low-signal alerts cause users to become numb to, and as a result, dismiss real warnings.

- **Informal learning & knowledge transfer** (Obj. 5), Narrative-based learning outperform formal training in changing behaviour of users.